

Clínica Universidad de Navarra

Principles of Laboratory Medicine

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Learning goals

- Interpretation of test results I
  - Reference intervals
  - Clinical decision limits
  - Critical results
  - Reference Change Value
- Interpretation of test results II
  - Sensitivity / specificity / diagnostic efficiency
  - ROC curves
  - Positive and negative predictive values

Interpretation of tests results I

1. Reference Interval: Range observed in 95% of a reference population
2. Clinical decision limits: Thresholds guiding diagnosis, treatment decisions..
3. Critical values: Immediate clinical action required
4. Reference Change Value: Interpretation of consecutive results in the same patient

1. Reference interval

BIOQUIMICA SANGUINEA Validación: Dr. Vallés Díez

Tipo de Muestra: **Suero**

Parámetro		*Resultado*	*Unidades*	*Valores de Referencia*
Glucosa		**85**	mg/dL	74 - 100
Urea		**32**	mg/dL	17 - 43
Creatinina		**0.85**	mg/dL	0.51 - 0.95
Filtrado glomerular estimado (CKD-EPI)		**78.5**		
		SEQC/SEN CKD-EPI:: FGE >60 mL/min/1,73 m^2 filtración probablemente normal. Un FGE < 60 mL/min/1,73 m^2 es indicador de enfermedad renal.		
Ácido úrico		**3.9**	mg/dL	2.6 - 6
Proteínas totales		**7.2**	g/dL	6.6 - 8.3
Albúmina		**4.19**	g/dL	3.5 - 5.2
Bilirrubina total	*	**1.39**	mg/dL	0.3 - 1.2
Sodio		**142**	mmol/L	136 - 146
Potasio		**4**	mmol/L	3.5 - 5.1
Cloruro		**104**	mmol/L	101 - 109
Calcio total		**9.9**	mg/dL	8.8 - 10.6
Fósforo		**3.74**	mg/dL	2.5 - 4.5
Alanina aminotransferasa (ALT)	*	**54**	UI/L	0 - 35
Aspartato aminotransferasa (AST)	*	**45**	UI/L	0 - 35
Fosfatasa alcalina (ALP)		**98**	UI/L	30 - 120
Gamma-glutamil transferasa (GGT)	*	**78**	UI/L	0 - 38
Lactato deshidrogenasa (LDH)		**181**	UI/L	0 - 248

Quantitative, units, reference interval, flags to highlight unusual results

1. Reference interval

Definition

Range of results originated from a reference population.

It is not the same as "normal":

- Reference population, not always healthy
- Can refer to a specific population, pediatric, newborn, women, trimester of pregnancy, etc.
- The term "normal" implies gaussian distribution and some analytes are not normally distributed

Application and Clinical Context Importance

- identify abnormal results and assess patient health contextually.
- Must be interpreted considering clinical context and preanalytical factors.

Variability Factors

Reference intervals vary by age, sex, and physiological conditions to fit the tested population accurately. Also, depend on the sample and method

Establishment Process

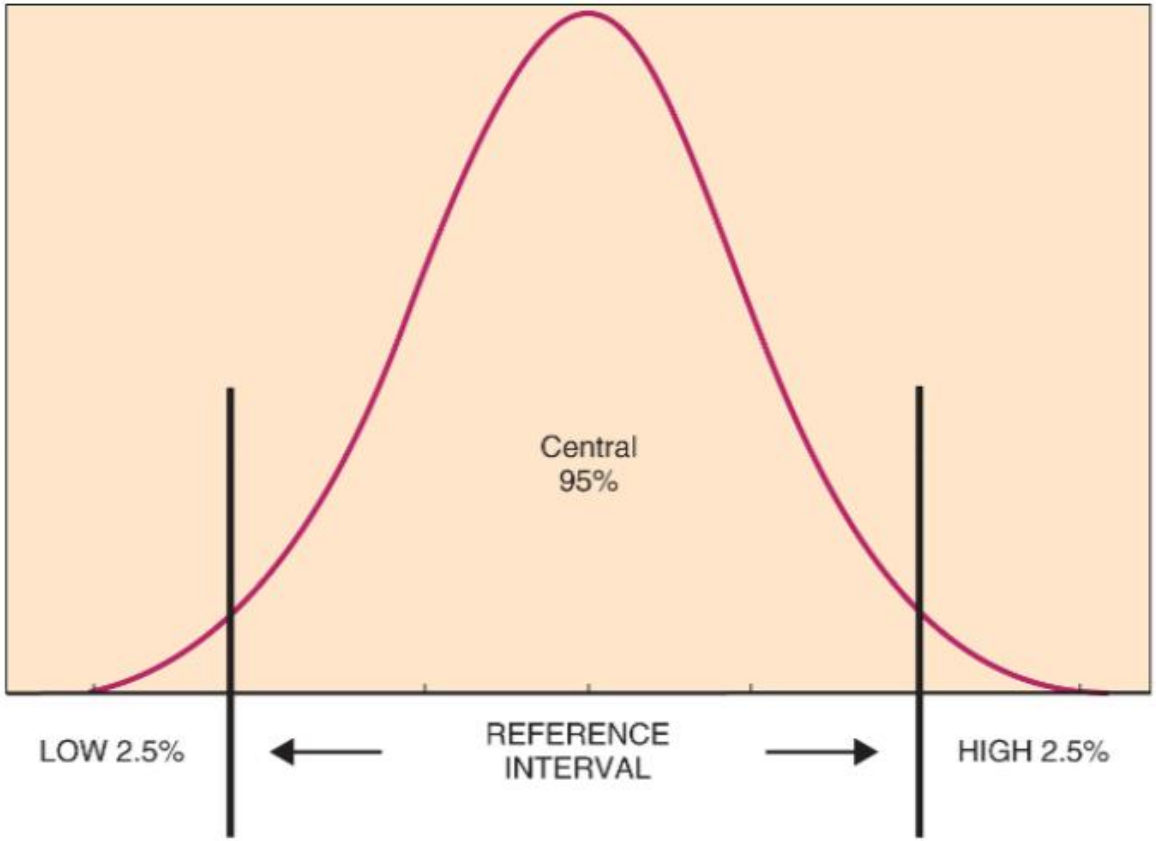
1, Generate own reference interval: Intervals typically cover the central 95% of values using statistical analysis of reference populations.

2, transference from elsewhere

Procedure to establish reference intervals

1. Generate own reference interval
   - Preanalytical phase
     - Select the reference population
       - Healthy / other groups (age, sex, relevant conditions)
       - Representative of the population
       - Available and limited number of selected individuals n=120
     - Identical patient preparation prior to sample collection
     - Standarized sample collection and processing
   - Analytical: method and sample

- Posanalytical
  - Eliminate outliers
  - Includes 95% of values the found in the reference population
  - Interval: P 2,5 y 97,5



2. Transference of reference intervals

Definition:

Using reference values from a validadted external source (another laboratory, manufacturer´s…) after verification

Need for transfer of reference values

Not affordable for all laboratories
limited resourses
Novel assays there are no other sources of data

Criteria for Transfer

Analytical methods should be comparable or harmonized.
Population characteristics should be similar.
Pre-analytical and analytical conditions must be similar.

Process of Transfer

Verification: Confirm that external reference values apply to the local population.
Ensure that the population, preanalytlical phase, sample and methods are similar
Testing a small number of individuals (usually 20.
Statistical comparison between local data and transferred RVs.

2. Clinical decision limits

Definition of Decision Limits

Thresholds that guide diagnosis and treatment based onclinical evidence and outcomes.

Difference from Reference Intervals

- Definition: reference ranges include 95% of the results of healthy people
- Generation:
    - Generated according to clinical criteria from clinical studies
    - Generated by consensus from national or international scientific societies
- Focus: on clinical significance (disease presence and severity) rather than normal population ranges.
- Highest in the hierarchy. In the laboratory report, clinical decision limits replace reference values

Clinical utility

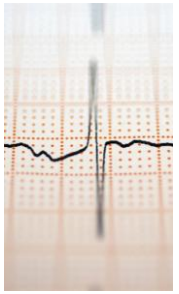
have a clinical association with specific diseases or risks

Example

Fasting glucose ≥126 mg/dL, diagnosis of diabetes. American Diabetes Association

Table 1. Terminology.	
Critical result	Results that are possibly life-threatening or lead to significant patient morbidity, necessitating prompt medical attention.
Significantly abnormal result	Results that, while not immediately life-threatening, demand timely medical attention and subsequent follow-up.
Critical test	Tests that necessitate rapid communication regardless of the test result.
Clinical decision limit	Thresholds that indicate a notable risk of adverse clinical outcomes or serve as diagnostic criteria for a particular disease, prompting the recommendation of a specific medical course of action.

3. Critical values



Definition
Critical values represent lab results significantly outside the reference range indicating life-threatening conditions requiring urgent clinical intervention.

Predefined Thresholds
Laboratories set predefined thresholds for critical values to standardize urgent alerts.
Defined by Bibliography + consensus with the clinicians

Urgent Communication Protocols
Critical values trigger immediate communication with the clinician,
Active/safe communication

Patient Safety Importance
Accurate identification, reporting and registration of critical values are vital for patient safety and care quality.

Examples of Critical values

TEST	MIN REF VALUE	MAX REF VALUE	MIN ALERT	MAX ALERT	UNITS
Pla-Sodium	136	145	120	160	mEq/L
Pla-Potasium	3,5	5,0	2,5	6,0	mEq/L
Pla-Glucose	74	100	40	400	mg/dL
Pla-Calcium	8,6	10,0	6,0	13,0	mg/dL

Role play

Lab tecnitian



Junior Doctor



Lab technician

Context

You work in the hospital's clinical laboratory. A blood sample from a patient in the Emergency Department has returned a **critical potassium value**.

The patient, Maria Lopez, is a 68-year-old woman admitted with **chest discomfort, fatigue, and mild confusion**. Blood was drawn 30 minutes ago. The potassium level is **6.9 mmol/L**, which is above the critical threshold. The sample has been reviewed and is **not hemolyzed**.

Aim

• Communicate the result to the physician encharged
•Confirm patient identity and sample validity.
•Registration

Tools

•Be clear and concise.
•Positive confirmation of identity and result

Junior doctor

Context

You are a first-year resident working in the Emergency Department. You are currently overwhelmed:

- You're reviewing multiple patient charts.
- You're trying to discharge two stable patients.
- A nurse just asked you to assess a new arrival with shortness of breath.
- You haven't had time to eat or take a break.

Suddenly, you receive a call from the lab about a critical potassium result.

Aim

- Listen carefully to the technician's
- Ask clarifying questions if needed (e.g., sample time, hemolysis, patient history).
- Confirm the patient's identity and result.
- Decide on next steps: assess the patient, repeat the test, start treatment, or escalate to a senior physician.
- Document the communication and your clinical plan.

Tools

- Positive confirmation of identity and result
- Stay calm

4. Interpretation of serial results

Reference change value (RCV)

Definition

RCV determines significant changes between consecutive lab results by considering variability factors (analytical and biological)

.

Clinical Relevance

RCV helps clinicians identify significant changes beyond variations due normal fluctuations in patients or in measurement methods

Calculation

$CV_T = [CV_P^2 + CV_A^2 + CV_B^2]^{1/2}$

Total Variation	Pre-analytical Variation	Analytical Variation	Biological Variation
	Minimised by standardisation of sample collection, transport and handling	Established by replicate analysis of quality control samples	Estimates of within-subject biological variation obtained from the published literature

$CV_T = (CV^2_A + CV^2_I)^{1/2}$

$RCV = 2^{1/2} \times Z \times CV_T = 2^{1/2} \times Z \times (CV^2_A + CV^2_I)^{1/2}$

Change reference value

Male age 63

Cholesterol at first visit: 260 mg/dL
Cholesterol after healthy habits: 240 mg/dL
RCV for a significant change (95%) = 17,2%

Is it a significant change?

1. Calculate the difference: 260-240= 20 mg/dL
2. Calculate the change (%) compared with the first value =(20/260)*100 = 7,6%
3. If the observed change is lower than the RCV, it is not significant

Clinical Case

A patient with gout has urate plasma levels of 7.5 mg/dL (++). He comes in for a check-up after 3 months of treatment with a new drug to reduce urate plasma levels. This time urate levels are 6.6 mg/dL. The reference change value for urate in the laboratory where the samples were analyzed is 13.8%. Is the change in the result significant?

It is not significant

It is significant

It depends on the positive predictive value

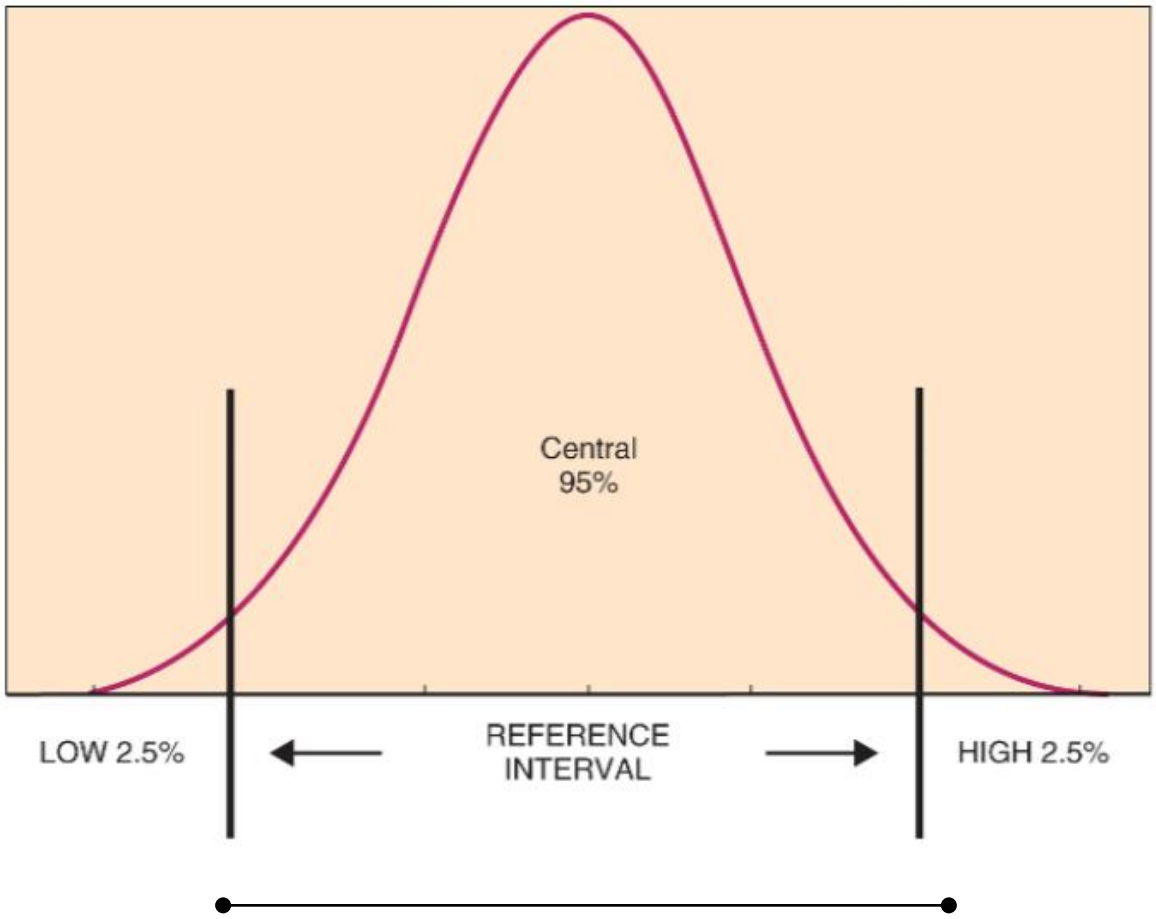
It will depend on the diagnostic sensitivity

Final exam nov 24

Interpretation of test results II

Analytical Performance

Reference interval



Includes 95% of the values found in the reference population

Discriminating capacity

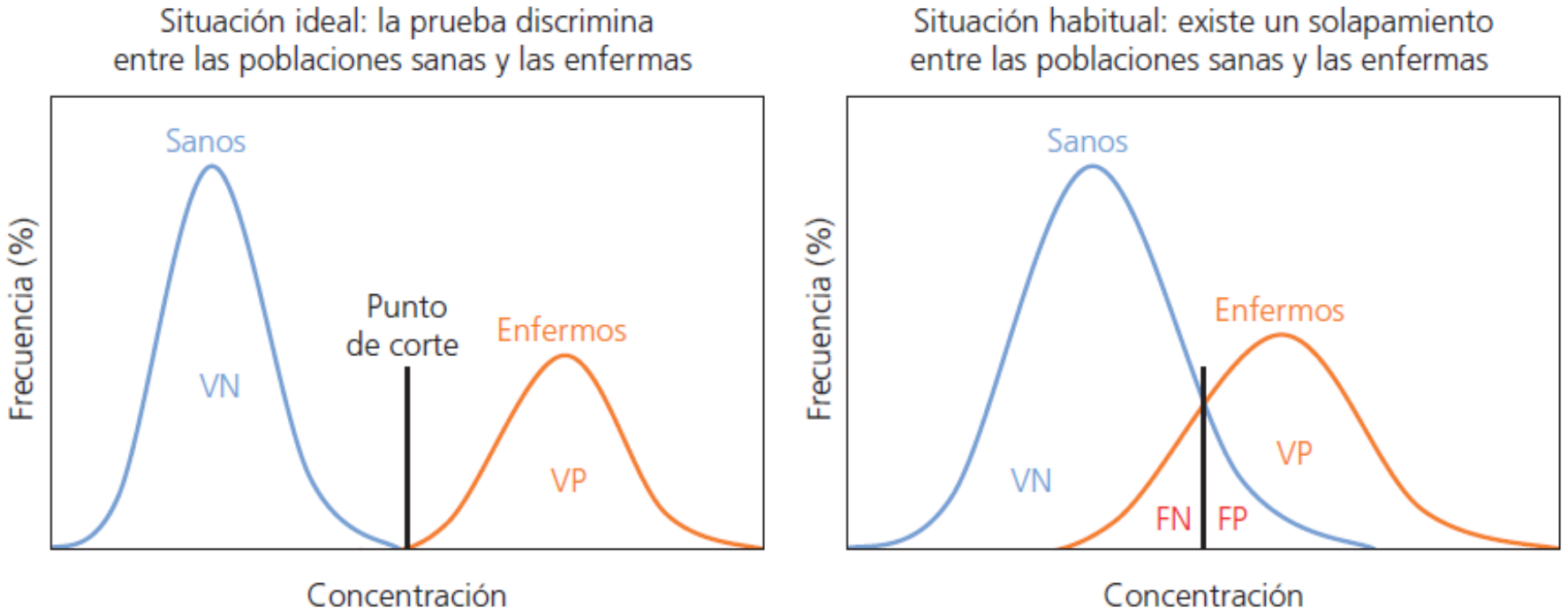
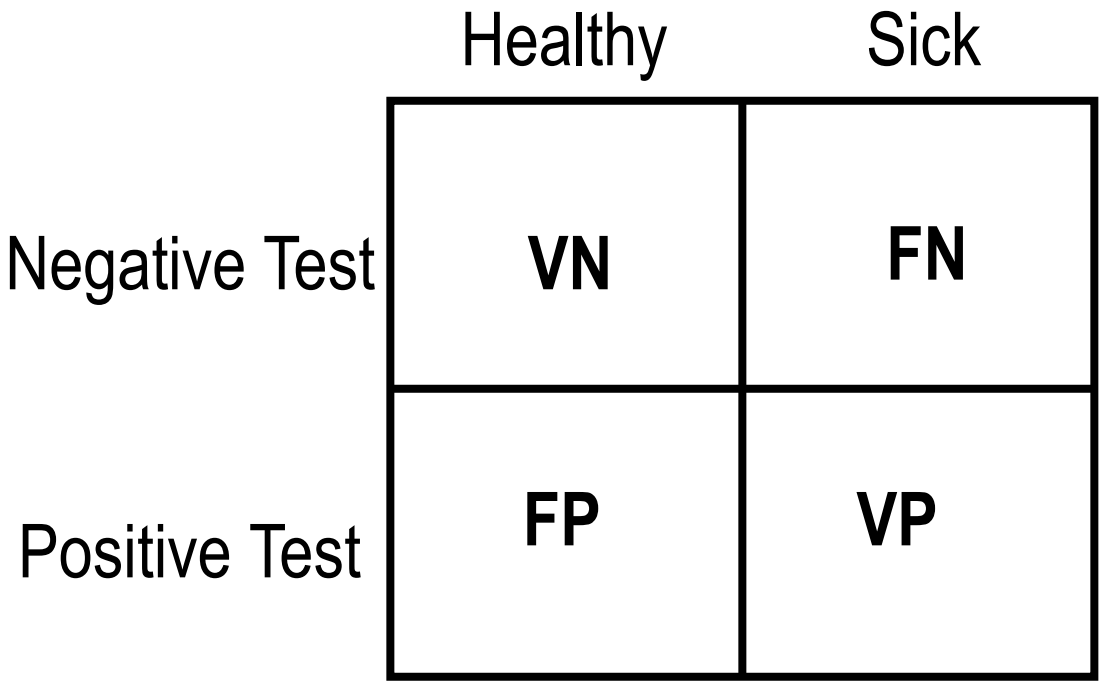


Figura 2-2 Solapamiento de resultados de una magnitud bioquímica entre la población normal y la población enferma con presencia de falsos positivos y negativos. FN: falso negativo; FP: falso positivo; VN: verdadero negativo; VP: verdadero positivo.

Double-entry table



Diagnostic Sensitivity and specificity

Diagnostic sensitivity

$$S = \frac{TP}{TP + FN} \times 100\ \%$$

Percentage of positive results (+) in the presence of disease
Percentage of sick people corrected clasified by the test

Diagnostic specifity

$$E = \frac{TN}{TN + FP} \times 100\ \%$$

Percentage of negative results (-) in the absence of disease
Percentage of healthy people corrected clasified by the test

Diagnostic efficiency

$$EF = \frac{TP + TN}{TP + TN + FP + FN} \times 100\ \%$$

Percentage of correct results
Percentage of people corrected clasified by the test

	Healthy	Sick
Negative Test	**496 (VN)**	**50 (FN)**
Positive Test	**4 (FP)**	**250 (VP)**

$$\text{Sensitivity} = \frac{TP}{TP + FN} \times 100\% = \frac{250}{250 + 50} \times 100\% = \mathbf{83\%}$$

$$\text{Specificity} = \frac{TN}{TN + FP} \times 100\% = \frac{496}{496 + 4} \times 100\% = \mathbf{99\%}$$

$$\text{Efficiency} = \frac{TN + TP}{VN + VP + FP + FN} \times 100\% = 93{,}2$$

Cut-off for clinical decisions

- Many parameters are not dichotomic, a cut-off is required.

- The cut-off determines the number of TP, TN, FP and FN.

The cut-off depends on the clinical decisions after test results

1) High sensitivity

Detect max number of sick people (↑ TP y ↓ FN)

a) Neonatal screening

b) Serious tratable disease: Infarction

2) High specificity

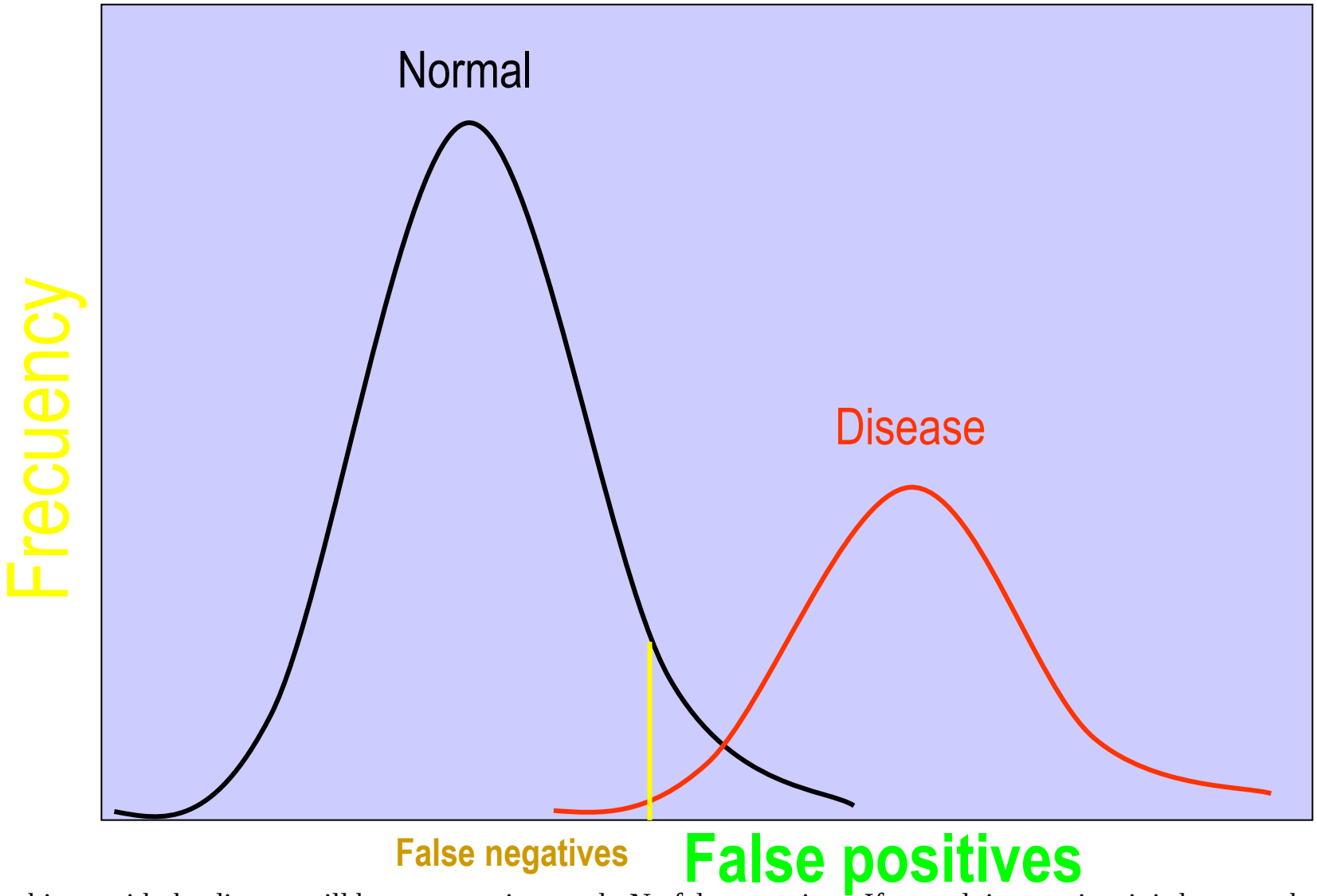
Detect healthy (↑ TN y ↓ FP)

a) Serious non-tratable disease. Avoid FP wronly diagnosed.

b) Treatment with high morbidity and no high efficacy

Screening tests	→	High sensitivity
Confirmatory tests	→	High sensitivity High Specificity

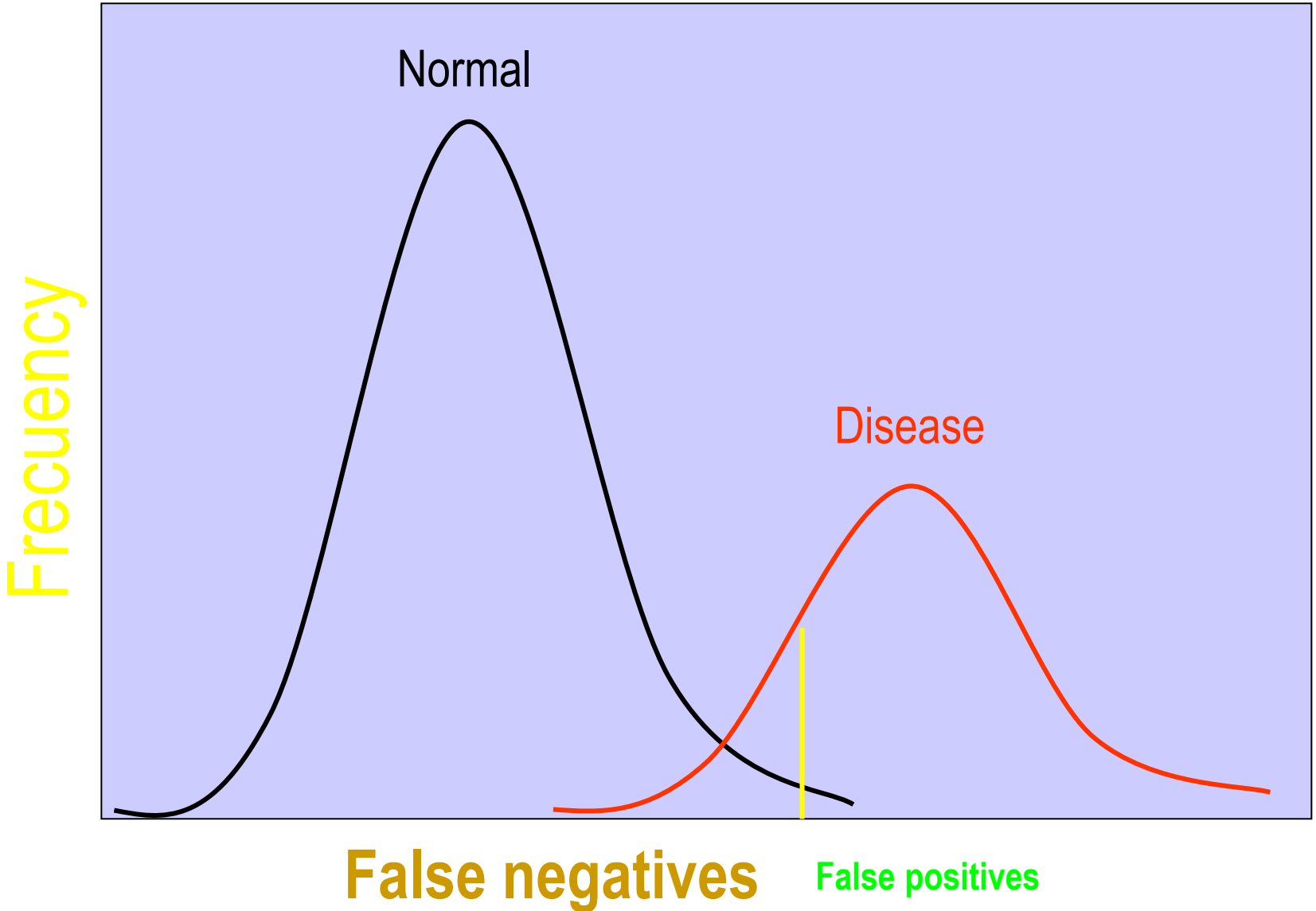
Setting the cut-off for high sensitivity



No subjects with the disease will have a negative result. No false negatives. If a result is negative, it is because the person is healthy

High sensitivity rules out the disease when the test is negative

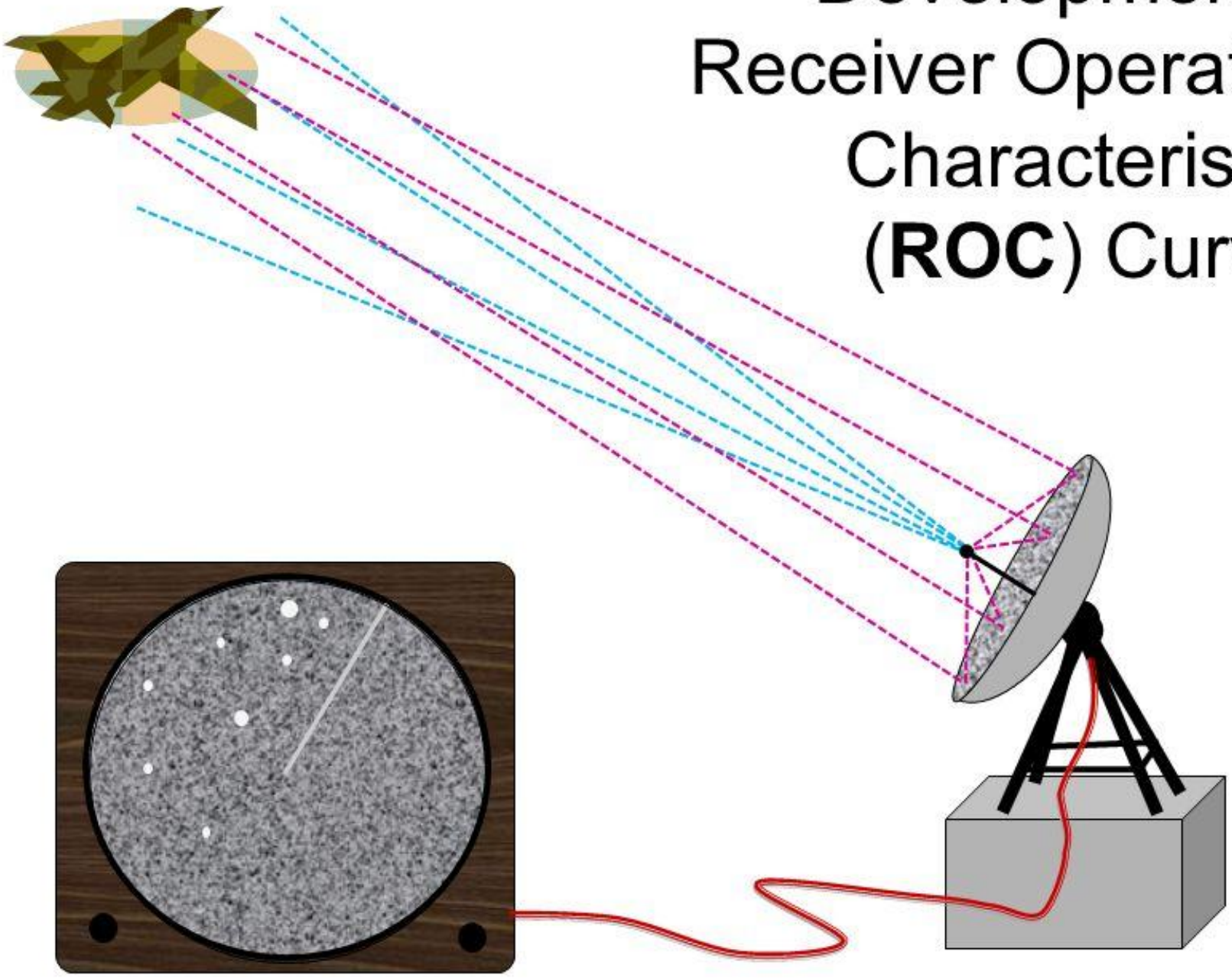
Setting the cut-off for high specificity



If E=100%. There are no false positives. When the test result is positive it is because the person has the disease.

A high specificiy rules in the disease when the test result is positive

Development of Receiver Operating Characteristics (**ROC**) Curves



ROC curves

Conc.	Sens.	1 – Esp.
0	1	1
3	0,96	0,8
5	0,91	0,5
8	0,88	0,37
12	0,65	0,37
18	0,48	0,12
30	0	0

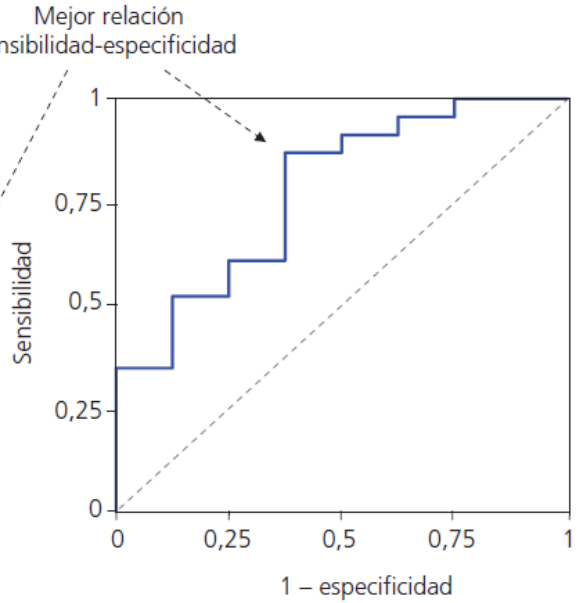


Figura 2-4 Curva ROC de una magnitud bioquímica.

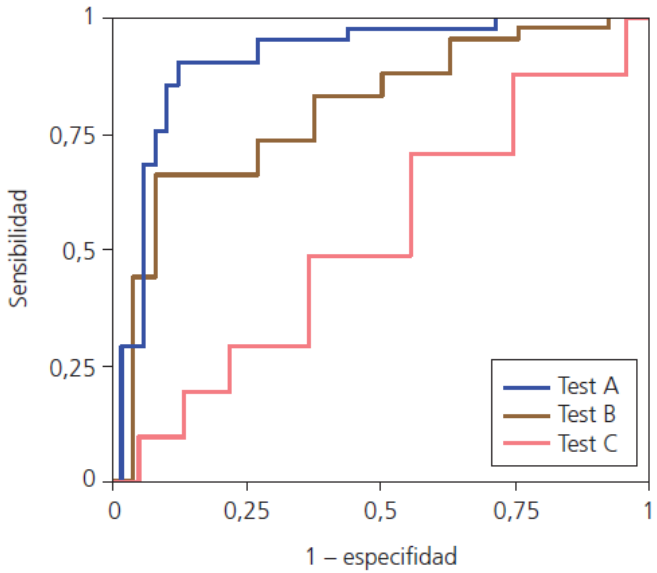


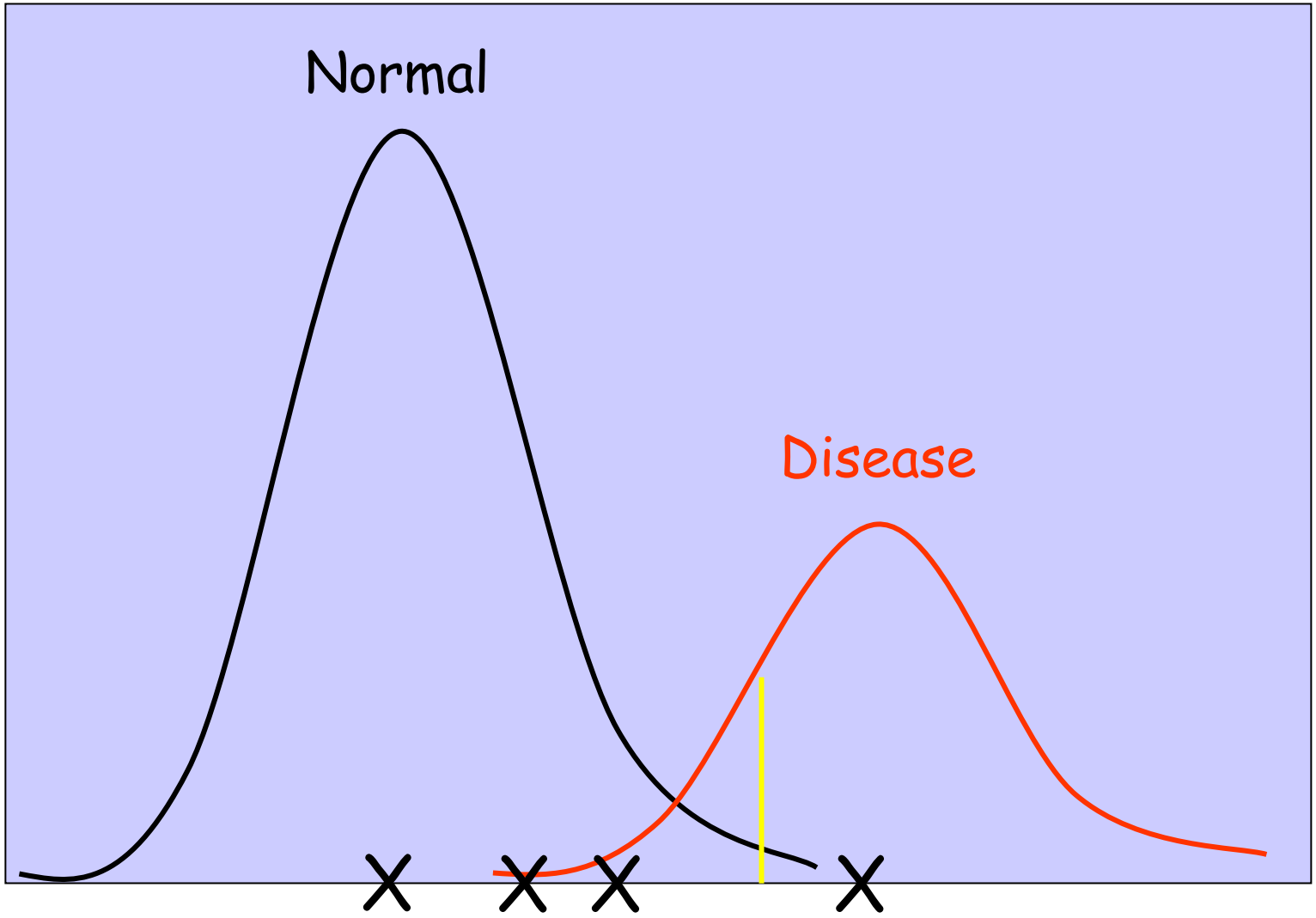
Figura 2-5 Selección de la eficiencia de un test en función del área bajo la curva de una representación ROC. El test A es el más eficiente, mientras que el C no tiene utilidad alguna.

AUC	Discriminating power of the test
0,9-1,0	Optimal
0,8-0,9	Good
0,7-0,8	Normal
0,6-0,7	Poor efficiency
0,5-0,6	None

ROC curves

- Show the association between sensitivity and specificity
- The closer to the top left, the more accurate the test is
- Closer to the diagonal, test less accurate
- The AUC reflects the test accuracy

Predictive value



Has someone the disease when the test result is X ?

Predictive values

Positive Predictive Value $PPV = \frac{TP}{TP + FP} \times 100\ \%$

Probability of having the disease when the test is (+)

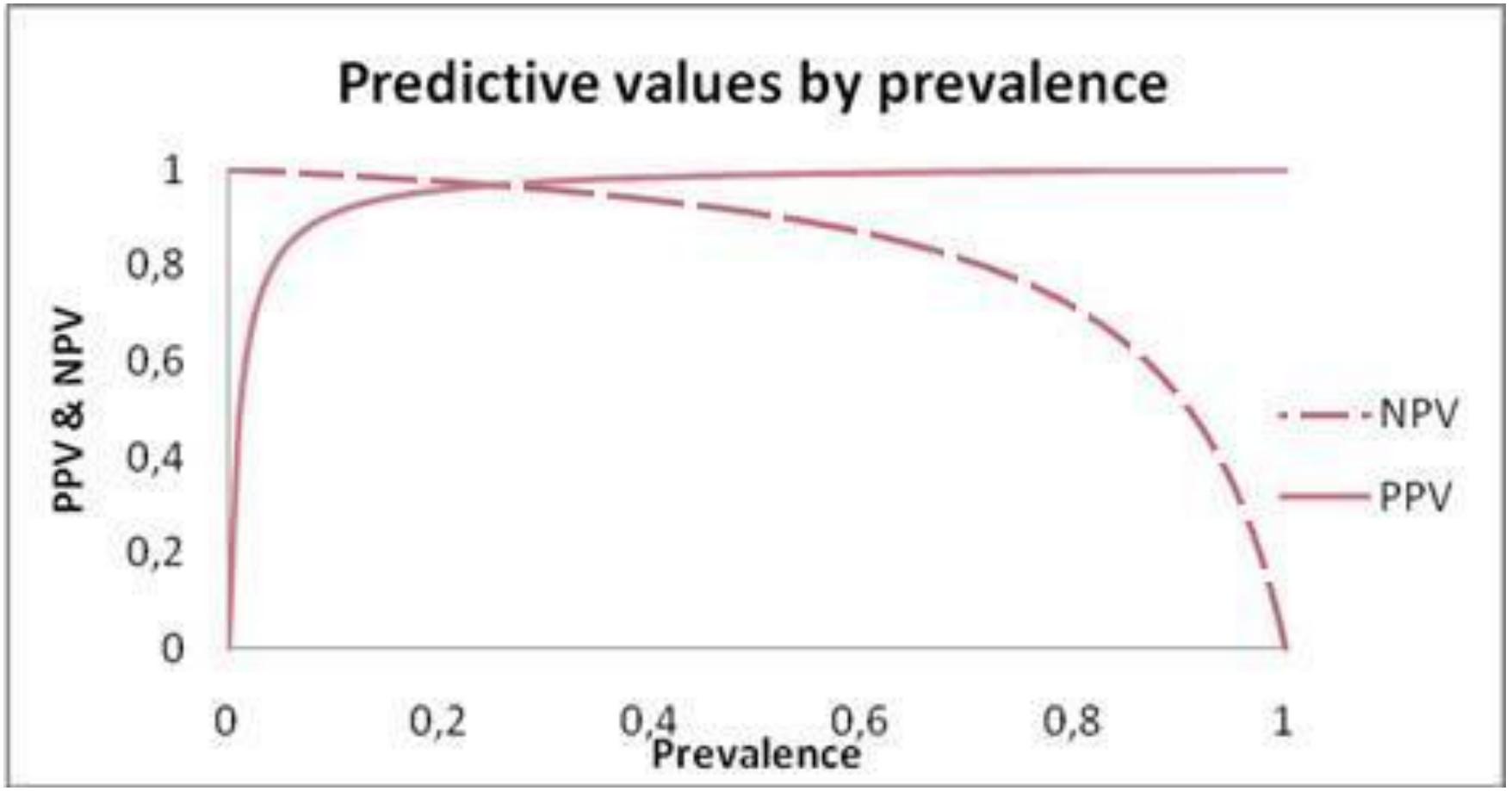
Negative Predictive Value $NPV = \frac{TN}{TN + FN} \times 100\ \%$

Probability of being healthy when the test is (-)

	Health	Disease
Negative test	**(TN)**	**(FN)**
Positive test	**(FP)**	**(TP)**

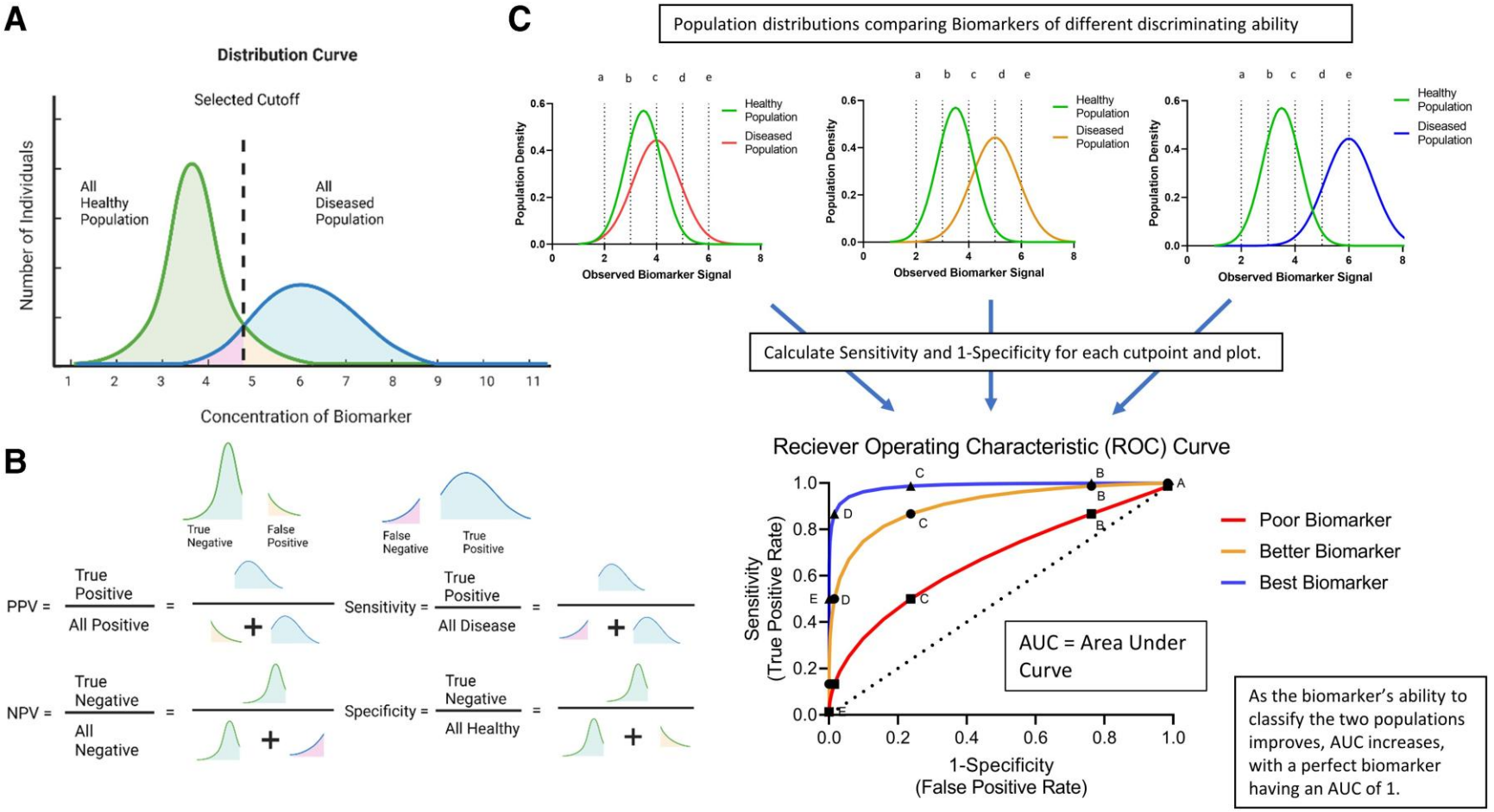
S and E are used to select a test

PPV and NPV are use for patients´results interpretation



↑ Prevalence	→	↑ PPV and ↓ NPV
↓ Prevalence	→	↓ PPV and ↑ NPV

Fig. 3. Biomarker performance characteristics. (A), Hypothetical probability density distributions for a biomarker ...



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Summary

1. The reference interval supplied with the test result gives only the probability of the results being ‘normal’ or ‘abnormal’.

2. Reference intervals may vary according to the age or sex of the patient or other factors.

3. An abnormal result does not always indicate that a disease is present, nor a normal result that it is not. Reference intervals are merely a guide to whether a patient has a disease.

4. If a result does not accord with that expected for the patient, the finding should be discussed with a professional in laboratory medicine and a repeat test arranged.

5. sensitivity is the proportion of patients with disease correctly identified by a test; specificity is the proportion of people without the disease correctly identified.

6. PPV is the proportion with a positive test who have the disease. NPV is the proportion of people without the disease with a negative result. [1 minus NPV] is the proportion with a negative test result who do have the disease.

7. A test is least useful when it confirms what you already know or strongly suspect and most useful when you do not know what the test result is likely to tell you.